

The Eidos

Specification and Deployment Guide

ARC-AGI-3 Interactive Reasoning Benchmark

SubThought Corporation
Version 1.0 — June 2026

The Eidos: Specification and Deployment Guide

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1 Overview

The Eidos is a Psyche: the interface between a GIL mind and the ARC-AGI-3 interactive reasoning benchmark. It mediates two flows — percepts traveling inward from the grid world to the mind's mechanisms, and actuations traveling outward from the mind's mechanisms to the grid world. Unlike the Aether (internet) and the Expanse (physical reality), the Eidos interfaces with a turn-based puzzle environment where the world changes only when the agent acts.

ARC-AGI-3 tests four capabilities that map directly to GIL's architecture: exploration (Perceiver), modeling (Totality), goal-setting (Coordination), and planning/execution (Executor). The world is a 64×64 grid with 16 colors and 7 standardized actions. Performance is scored by RHAE: $(\text{human_actions} / \text{ai_actions})^2$.

The Eidos is pure Premise — no DLL, no bridge adapter. Premise's built-in encode/decode json and native HTTPS tell/ask talk directly to the ARC-AGI-3 REST API.

2 The Percept Relation

The Percept relation from totality.theory is canonical. For the Eidos, all percepts carry :Modality Eidos and :Channel Grid.

Slot	Meaning	Eidos Examples
:Modality	Which Psyche.	Eidos (always)
:Channel	Sub-channel.	Grid (always)
:Address	API URL.	"https://three.arcprize.org"
:Data	Percept type.	grid-state, grid-delta, game-event
:Content	Structured data idiom.	{:GameId "Is20" :Grid {...} :State playing ...}
:Moment	When received.	\@m{2026060100000000}

3 Percepts

3.1 grid-state

Primary percept. Delivered after every action. Contains the full grid in the configured format, game state, level progress, action count, and available actions.

Slot	Type	Description
:GameId	string	ARC-AGI-3 game identifier (e.g., "Is20")
:Level	integer	Current level within the game
:Width / :Height	integer	Grid dimensions — always 64
:Format	literal	Grid encoding: raw, sparse, or rle
:Grid	list	Grid data in the configured format
:State	literal	playing, win, or game-over
:LevelsCompleted	integer	Levels solved so far
:TotalLevels	integer	Total levels in this game
:ActionsUsed	integer	Agent's action count
:HumanActions	integer	Human baseline action count
:AvailableActions	list	Valid actions for current state

3.2 grid-delta

Emitted alongside grid-state when format is raw. Shows only what changed: list of { :I index :From old :To new } entries.

3.3 game-event

Emitted when the game state transitions to win or game-over. Contains :State, :LevelsCompleted, :TotalLevels.

4 Actuations

Seven standardized actions. The grid changes only when the agent acts.

:Action	ARC Name	Description	:Parameters
reset	RESET	Start or restart game/level	{:Gameld "Is20"} or {}
up / action1	ACTION1	Simple — up	{}
down / action2	ACTION2	Simple — down	{}
left / action3	ACTION3	Simple — left	{}
right / action4	ACTION4	Simple — right	{}
interact / action5	ACTION5	Simple — interact/select	{}
click / action6	ACTION6	Complex — x,y coordinates	{:X 32 :Y 16}
undo / action7	ACTION7	Undo last action	{}

5 Cross-Cutting Structure

```

PERCEPT: [PERCEPT :Modality Eidos :Channel Grid :Data D :Content {...} :Moment M
:Token T]
ATTEMPT: [ATTEMPT :Action A :Parameters {...} :Token T :By M]
RESULT: [RESULT :Action A :Status S :Reason R :Moment M :Token T]
URGE: [URGE :Need N :Source S :Delta D :Moment M :Token T]

```

Every action produces exactly one grid-state percept. Optionally also a grid-delta (raw format, if grid changed) and a game-event (if state changed to win or game-over).

5.1 Urges

Need	Source	Trigger
connectivity	Grid	API connection lost
efficiency	Grid	Action count exceeds 3× human baseline (RHAЕ < 0.11)

6 Grid Representations

Raw:

Flat list of 4096 integers. Row: $y = \text{index} / 64$. Column: $x = \text{index} \bmod 64$. Simplest, no processing, full spatial data.

Sparse:

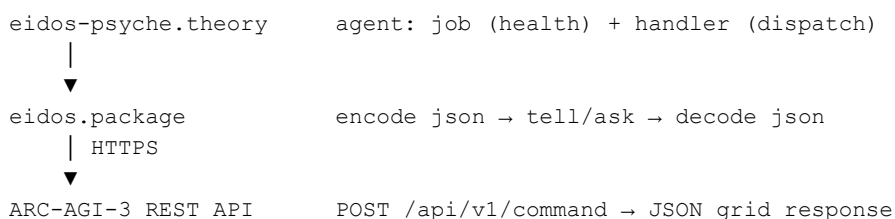
List of $\{I \text{ index} : C \text{ color}\}$ for non-zero cells. Compact for mostly-empty grids. 30 colored cells = 30 pairs instead of 4096 values.

Run-Length:

Pairs of count and color: $\{64 \ 0 \ 3 \ 3 \ 59 \ 0 \ \dots\}$. Compact for large uniform regions. Preserves scan order.

Utility functions: grid-to-sparse, grid-to-rle, sparse-to-grid, rle-to-grid. Also: grid-cell (x,y lookup), grid-row, grid-diff, grid-colors, grid-count.

7 Architecture



No DLL, no provider, no bridge. Pure Premise talks directly to the ARC REST API.

8 Deployment

File	Description
eidos.package	Pure Premise. API calls, actions, grid compression, utilities, scorecards
eidos-psyche.theory	Single agent: minimal job + turn-based handler
Eidos.daicho	API URL, API key, grid format, urge thresholds
eidos.suites	7 test suites: compression, utilities, connection, actions, percepts, scorecards, lifecycle

9 API Key

The ARC-AGI-3 API key is stored in $[API :Key]$ in the Eidos.daicho. Obtain it at <https://three.arcprize.org/user>. The key serves two purposes: HTTP Authorization header for every API request, and device credential registered with the GIL Registrar (the $:Token$ on every tuple).

10 Etymology

Eidos (Greek εἶδος, *eidos*) — from the verb εἶδω (*eidō*), “to see.” In Greek, seeing and knowing are the same word. The perfect tense of *eidō* is *oīda* — literally “I have seen,” but meaning “I know.” Plato elevated *eidos* to mean the ideal Form — the unchanging reality behind appearances.

Every ARC-AGI-3 puzzle asks: what is the *eidos* behind these colored tiles? The surface is a 64×64 grid of colors. The *eidos* is the transformation rule. Finding the *eidos* is solving the puzzle.

Where *Itē* names what’s at stake (the throne), *Aether* names the medium (propagation), and *Expanse* names the territory (vastness), *Eidos* names the act of perception itself — the moment where raw sensation becomes understanding.

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